

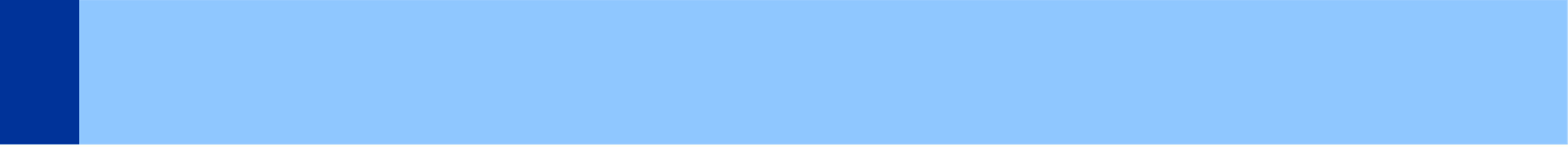
Biostatistics

By: Awgichew Kifle

Email: awgistat@gmail.com

Mobile: 0906813229

June, 2023



Estimation and hypothesis testing

Objectives

At the end of this course students will be able to:

- Define Inferential statistics
- Know statistical estimation
- Understand hypothesis testing & the “types of errors” in decision making.
- Use test statistics to examine hypothesis about population parameter

Inference



Use a random sample to learn something about a larger population

Inferential Statistics

- **Inferential Statistics:** Are **statistical methods used** for drawing conclusions about a **population** based on the information obtained from the a **sample** of observations drawn from that population

Inferential Statistics

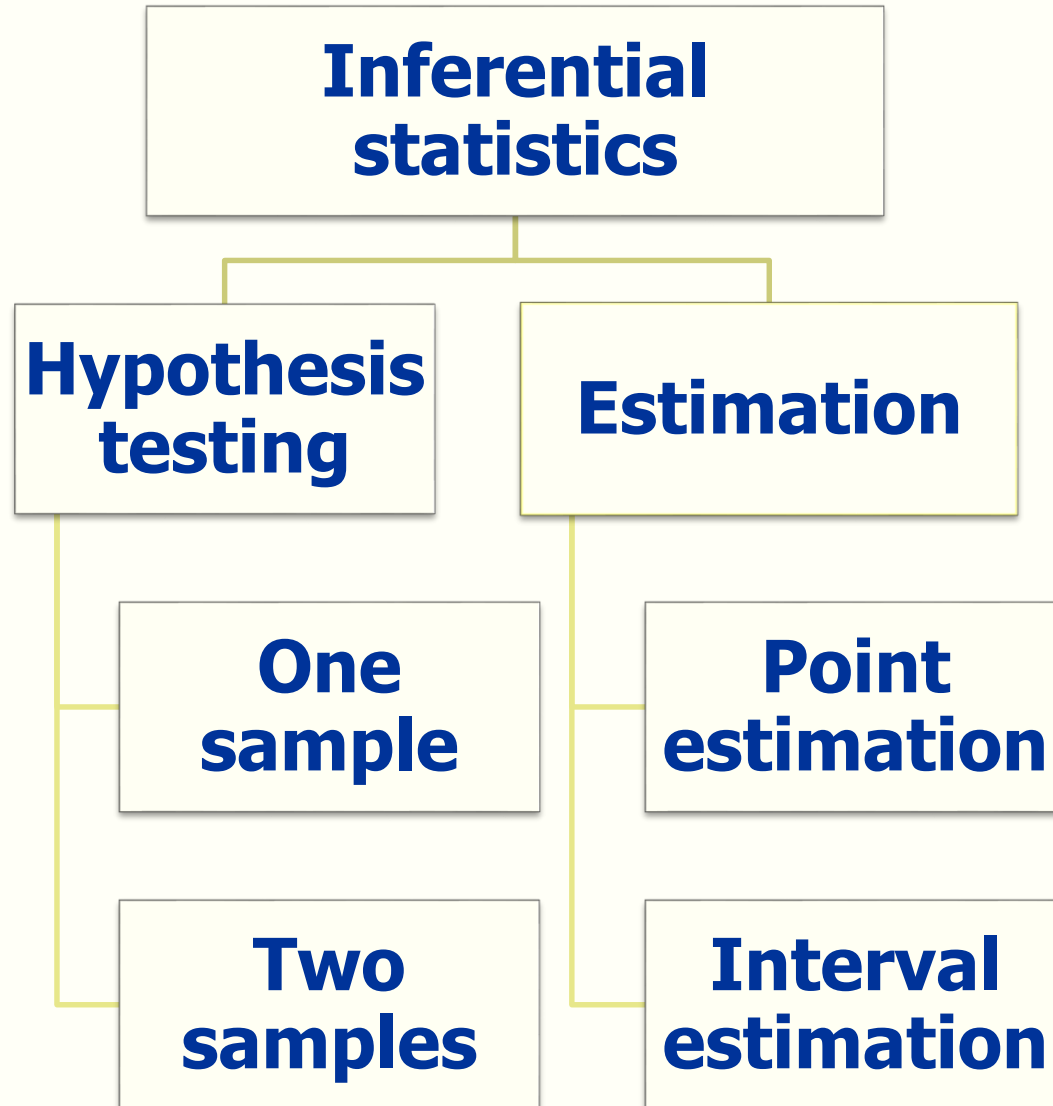
- Involves
 - Estimation
 - Hypothesis testing
- Purpose
 - Make decisions about population characteristics



Population?



Inferential Statistics



Inferential process

**Estimate
& test
population
parameter**



Population



**Sample
statistic**



Sample



Statistical Estimation

- Estimation is the process of determining a likely value of population parameter, based on information collected from the sample.
- Estimation is the use of sample statistics to estimate the corresponding population parameters
- The objective of estimation is to determine the ***approximate value*** of unknown population parameter on the basis of a sample statistic

- ◆ We assumed that the number of neutrophils in a sample of 100 WBC was binomially distributed with parameter $\mu=2$.
- ◆ We assumed that the number of bacteria colonies on 100 cm² agar plate was Poisson distributed with parameter $\mu=2$.
- ◆ We assumed that the distribution diastolic blood pressure measurement in 35-44 year old men was normal with mean $\mu=80$ mm Hg.

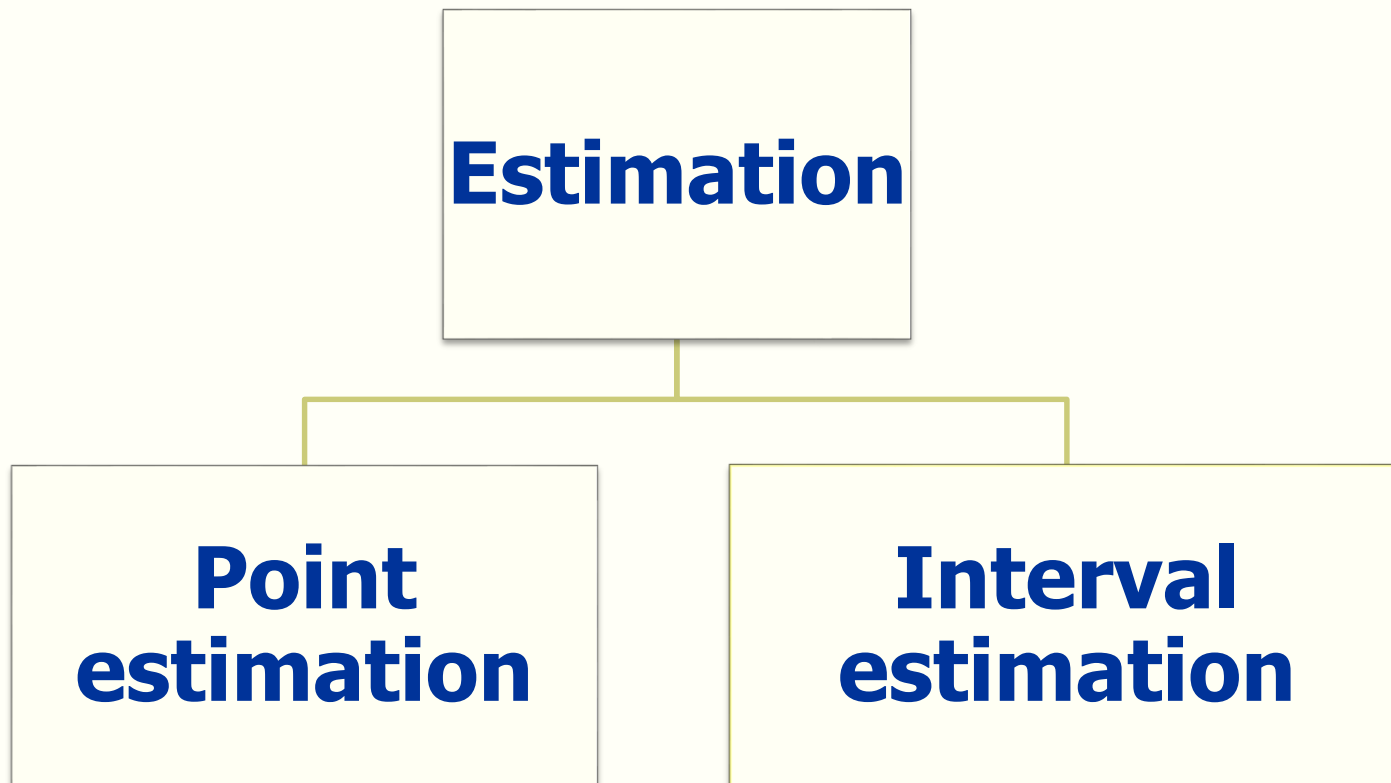
Sample Statistics as Estimators of Population Parameters

A sample statistic is a numerical measure of a summary characteristic of a sample.

A population parameter is a numerical measure of a summary characteristic of a population.

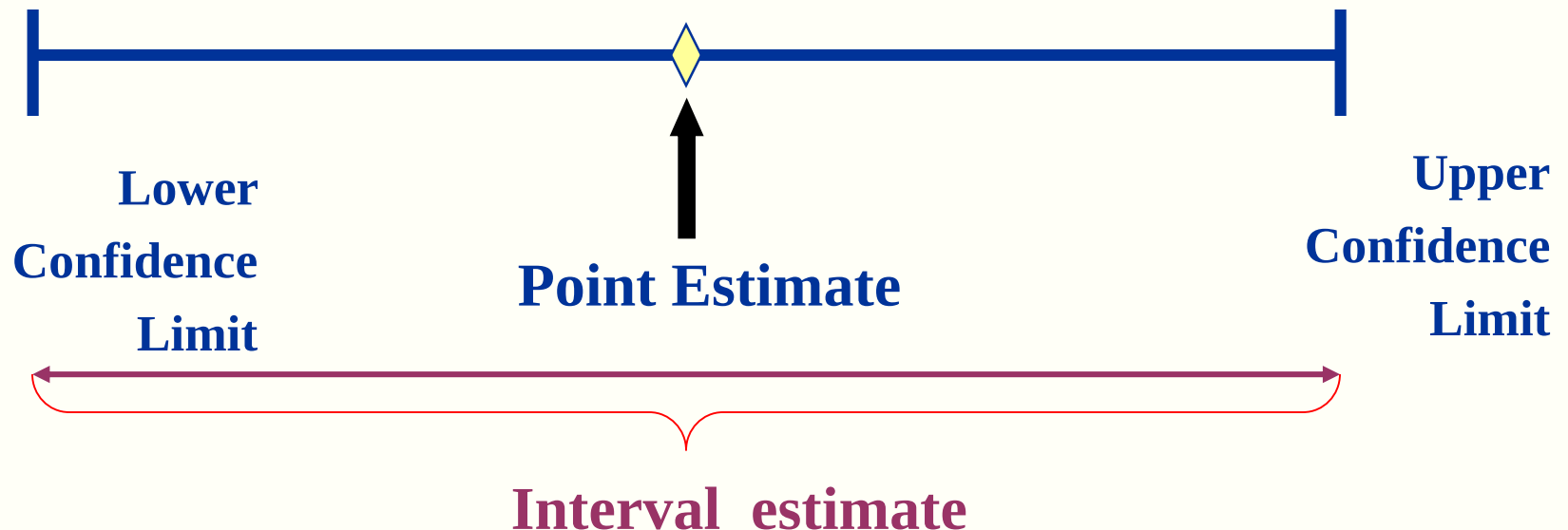
- An **estimator** of a population parameter is a sample statistic used to estimate or predict the population parameter
- An **estimate** of a parameter is a *particular* numerical value of a sample statistic obtained through sampling.

Estimation



Point and Interval Estimates

- A **point estimate** is a single value used as an estimate of a population parameter
- **Interval estimate** is a range or interval of numbers believed to include unknown population parameter with a certain degree of assurance
- *Point estimate is always within the interval estimate*



Estimation Process

Population

Mean, μ , is
unknown

Random
Sample

Point estimate

Mean

$$\bar{X} = 50$$

Interval estimate

I am 95%
confident that μ
is between 40 &
60.



Point estimation

- A single numerical value used to estimate the corresponding population parameter
- Gives little information about how close the value is to the unknown population parameter
- Example: Sample mean $\bar{X} = 3$ is **point estimate** of unknown population mean

Sample statistic & their corresponding population parameter

Statistic

Parameter

Mean:	$\bar{X}$	estimate	μ
Variance	s^2	estimate	σ^2
Standard deviation	s	estimate	σ
Proportion	p	estimate	π

From sample

From entire population

Properties of good estimate

a) Unbiasedness: An estimator is said to be **unbiased** if its expected value is equal to the population parameter it estimates.

- For example: when $E(\bar{X}) = \mu$, the sample mean is an unbiased estimator of the population mean
- Unbiasedness is an average or long-run property. The mean of any single sample will probably not equal to the population mean, but the average of the means of repeated independent samples from a population will equal to the population mean.

Properties of good estimate

b) **Minimum variance: (Efficiency)**

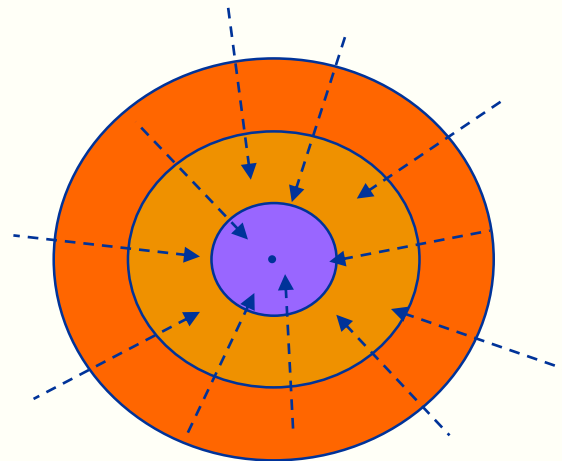
An estimate which has a minimum standard error is a good estimator

- For symmetrical distribution the mean has a minimum standard error and
- If the distribution is skewed the median has a minimum standard error

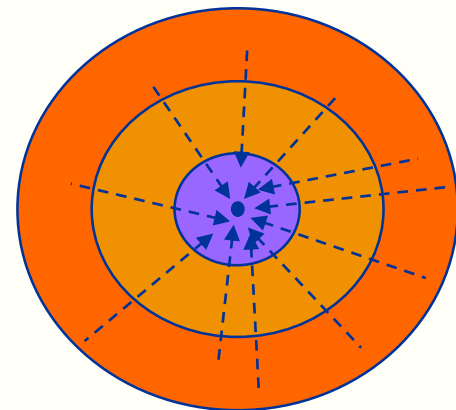
Properties of good estimate

C) Consistency: An estimator is said to be **consistent** if its probability of being close to the parameter it estimates increases as the sample size increases

Consistency



$n = 10$



$n = 100$

Interval estimation

- A single-valued estimate conveys little information about the actual value of the population parameter, about the accuracy of the estimate
- The probability of getting a sample statistic value that is exactly equal to the corresponding population parameter is usually quite small

Interval estimation

- It is not reasonable to assume that a sample statistic value is exactly equal to the corresponding population parameter
- An interval estimate which locates the population parameter within an interval, with a level of confidence is needed

Marginal Error

- A value calculated from the sample is the best guess when estimating corresponding population value
- Estimate is still uncertain due to sampling error
- Marginal Error is a measure of uncertainty
- Using marginal error you can state confidence interval: estimate $\pm$ marginal error

Confidence Interval or Interval Estimate

- ***Confidence interval*** or ***interval estimate*** is a range or interval of numbers believed to include an unknown population parameter
 - Associated with the interval, there is a measure of the ***confidence*** we have that the interval does indeed contain the parameter of interest
 - **Confidence interval:** provide a range of values of the estimate likely to include the “true” population parameter with a given probability
-
- A confidence interval or interval estimate has two components:
 - ✓ A range or interval of values
 - ✓ An associated ***level of confidence***

Confidence Level

1. Probability that the unknown population parameter falls within interval
2. Denoted $(1 - \alpha)\%$
 - α is probability that parameter is **not** within interval
3. Typical values are 99%, 95%, 90%

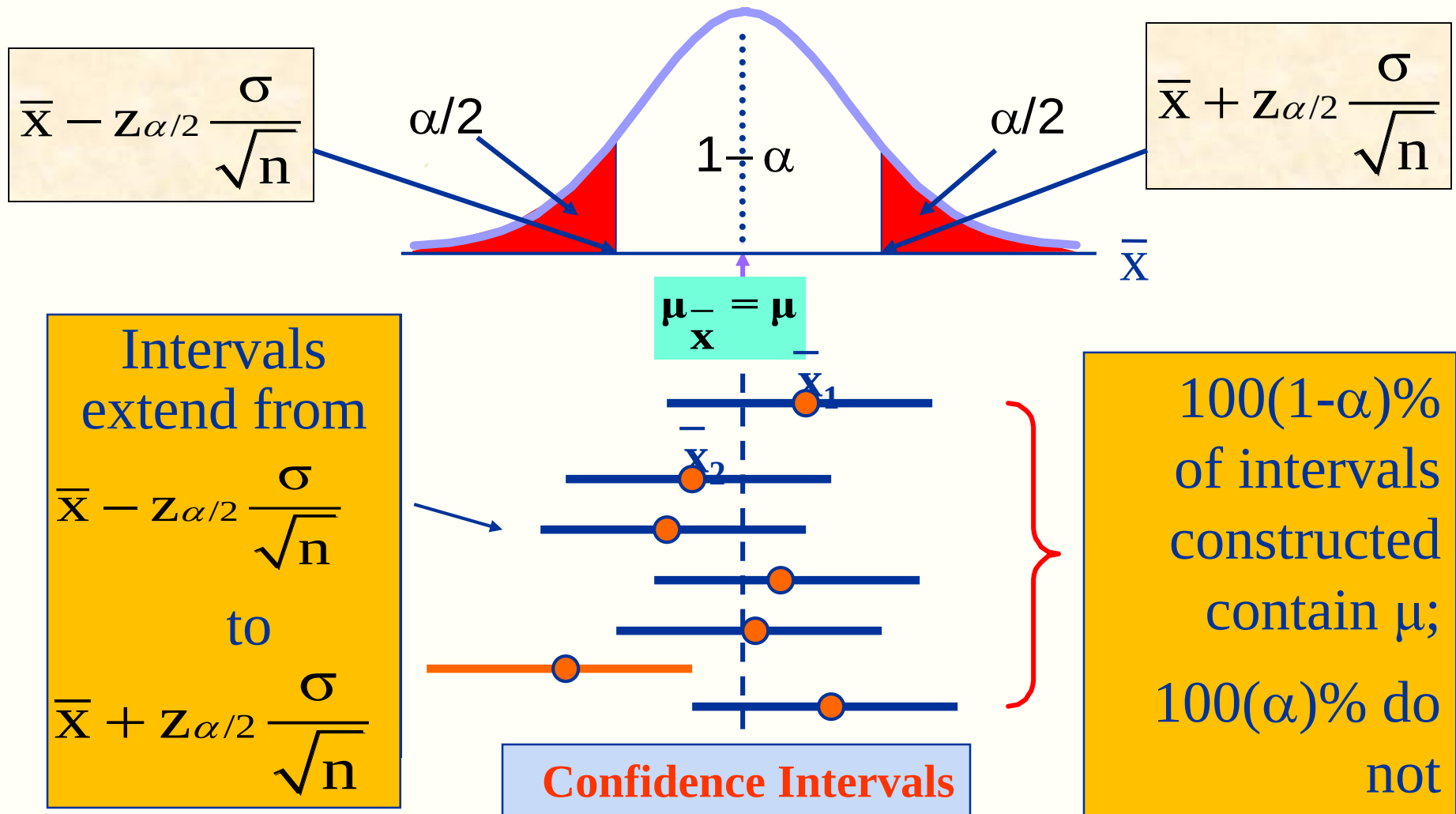
Common Confidence Levels

- Commonly used confidence levels are 90%, 95%, and 99%

<i>Confidence Level</i>	<i>Confidence Coefficient, $1 - \alpha$</i>	<i>$Z_{\alpha/2}$ value</i>
80%	.80	1.28
90%	.90	1.645
95%	.95	1.96
98%	.98	2.33
99%	.99	2.58
99.8%	.998	3.08
99.9%	.999	3.27

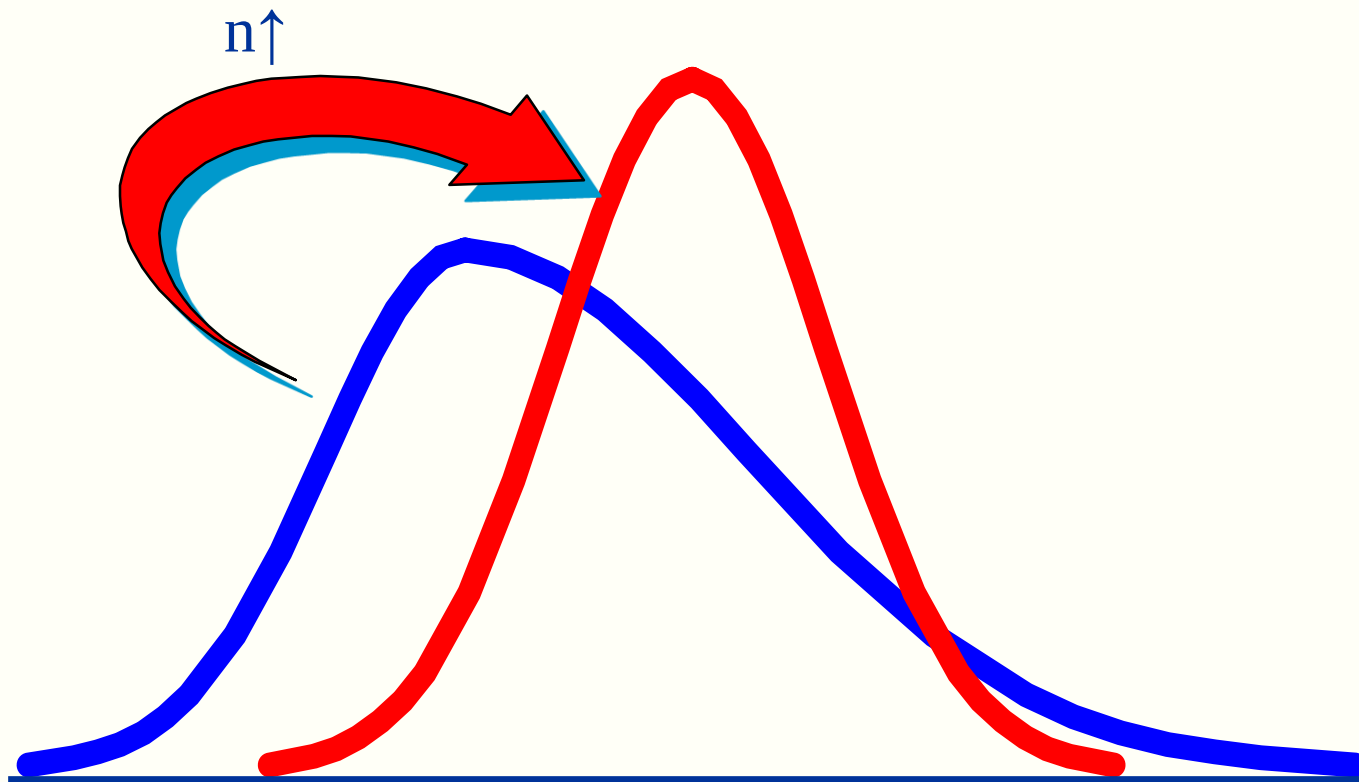
Intervals and Level of Confidence

Sampling Distribution of the Mean



Central Limit Theorem

- As the sample size gets large enough, the sampling distribution becomes almost normal regardless of shape of population



Central Limit Theorem

- **Rule of thumb:** Normal approximation for the sampling distribution will be good if $n \geq 30$. If $n < 30$, the approximation is only good if the population from which you are sampling is not too different from normal.
- Otherwise t-distribution

CI for population mean: Large-sample size and **when σ is known**

- For sufficiently large sample size $n \geq 30$, the sampling distribution of the sample mean, is approximately normal
- A $100(1-\alpha)\%$ C.I. for μ is:

$$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \Rightarrow \left(\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right)$$

- α is to be chosen by the researcher, most common values of α are **0.05**, 0.01, 0.001 and 0.1

CI for population mean: Large-sample size and **when σ is unknown**

- Whenever σ is not known (and the population is assumed normal), the correct distribution to use is the t distribution with $n-1$ degrees of freedom. **However,** for large degrees of freedom, the t distribution is approximated well by the Z distribution
- A large sample $100(1-\alpha)\%$ C.I. for μ is:

$$\bar{X} \pm z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

- **Note that:** when σ is unknown, s is a good approximation of σ

Example

- An epidemiologist studied the blood glucose level of a random sample of 100 patients. The mean was 170, with a SD of 10. Construct the 95% CI for the population mean.

Solution

$$\bar{X} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$170 \pm 1.96 \frac{10}{\sqrt{100}}$$

$$\Rightarrow (168.04, 171.96)$$

- We are 95% sure that the mean blood glucose level of the population lies between 168.04 and 171.96

CI for population mean: small sample size ($n < 30$) and when σ is unknown

- If population standard deviation is unknown, then the sample means from samples of size n are t-distributed with $n-1$ degrees of freedom
- A $100(1-\alpha)\%$ C.I. for μ is:

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

Example

- A study of hypoxemia during the immediate post-operative period reported the fractions of ideal weight for 11 patients who became severely hypoxemic during transfer to the recovery room. The mean is 1.51 and the standard deviation is 0.33. Estimate the 95% CI for the population mean fraction of ideal weight, where the population consists of hypoxemic patients similar to those in the study (use $\alpha=0.05$)

Solution

$$t_{\alpha/2, n-1} = t_{0.025, 10} = 2.2281$$

$$1.51 \pm 2.2281 \frac{0.33}{\sqrt{11}}$$

$$1.51 \pm 0.221$$

$$(1.289, 1.731)$$

- We are 95% sure that the population mean fraction of ideal weight μ lies between 1.289 and 1.731

The t Table

df	t0.100	t0.050	t0.025	t0.010	t0.005
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

CI for a population proportion: Large-sample size

- For sufficiently large samples, the sampling distribution of the proportion p is approximately normal
- A $100(1-\alpha)\%$ CI for π is:

$$p \pm Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

A sample is considered large enough when both $n \cdot p$ and $n \cdot q$ are greater than 5, where $q = 1 - p$.

Example

- A study on dental health practice. Of 300 adults interviewed, 123 said that, they regularly had a dental check-up twice a year. What is the 95% C.I. for π ?

$P = 123/300 = 0.41$ a point estimator of π

$$\alpha = 0.05 \Rightarrow Z_{0.025} = 1.96$$

$$p \pm Z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}}$$

$$0.41 \pm 1.96 \sqrt{\frac{(0.41)(0.59)}{300}}$$

$$\Rightarrow (0.3543, 0.4656)$$

Interpretation: ?

Thank you